

LFA Lista A1 A2

Engenharia de Computação
IFFluminense Campus Bom Jesus do Itabapoana
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Linguagens Formais e Autômatos
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PROVA A1:

Questões/Gabarito:

1. Cadeias que não contenham subcadeia abc

LHS		RHS
S	→	bS
A	→	aA
A	→	cS
A	→	bB
A	→	λ
S	→	cS
S	→	λ
S	→	aS
B	→	aA
B	→	bB
B	→	λ

2. Cadeias que contenham no mínimo dois símbolos a ou no máximo três símbolos c

LHS		RHS
S	$\rightarrow$	AaAaA
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	λ
A	$\rightarrow$	cA
S	$\rightarrow$	BcBcBcB
B	$\rightarrow$	aS
B	$\rightarrow$	bS
B	$\rightarrow$	λ

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3. Cadeias que iniciam e terminam com símbolos diferentes

LHS		RHS
S	$\rightarrow$	aAb
S	$\rightarrow$	aAc
S	$\rightarrow$	bAc
S	$\rightarrow$	bAa
S	$\rightarrow$	cAa
S	$\rightarrow$	cAb
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ

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4. cadeias que contenham pelo menos 3 símbolos iguais consecutivos

LHS		RHS
S	→	ABA
A	→	aA
A	→	bA
A	→	cA
A	→	λ
B	→	aaa
B	→	bbb
B	→	ccc

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5. Cadeias com quantidade par de símbolos a e impar de símbolos c
Gramática:

LHS		RHS
S	$\rightarrow$	aB
S	$\rightarrow$	bS
S	$\rightarrow$	cA
A	$\rightarrow$	λ
A	$\rightarrow$	aC
A	$\rightarrow$	bA
A	$\rightarrow$	cS
B	$\rightarrow$	aS
B	$\rightarrow$	bB
B	$\rightarrow$	cC
C	$\rightarrow$	aA
C	$\rightarrow$	bC
C	$\rightarrow$	cB

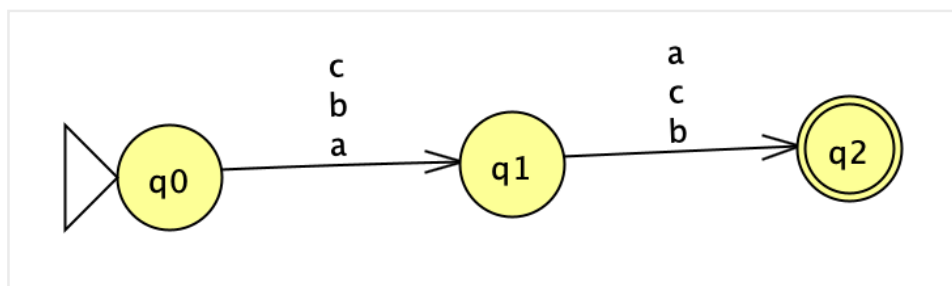
Prova A2:

Questões/Gabarito:

1. Cadeias de comprimento 2

ER: **(a|b|c)(a|b|c)**

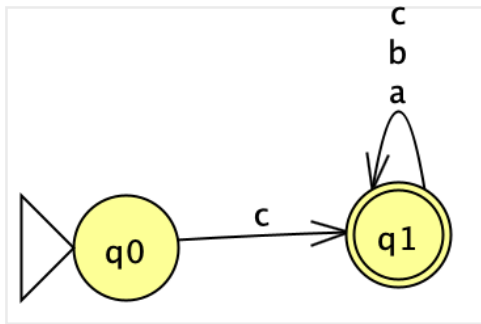
AF:



2. Cadeias que iniciam com c

ER: **c(a|b|c)***

AF:

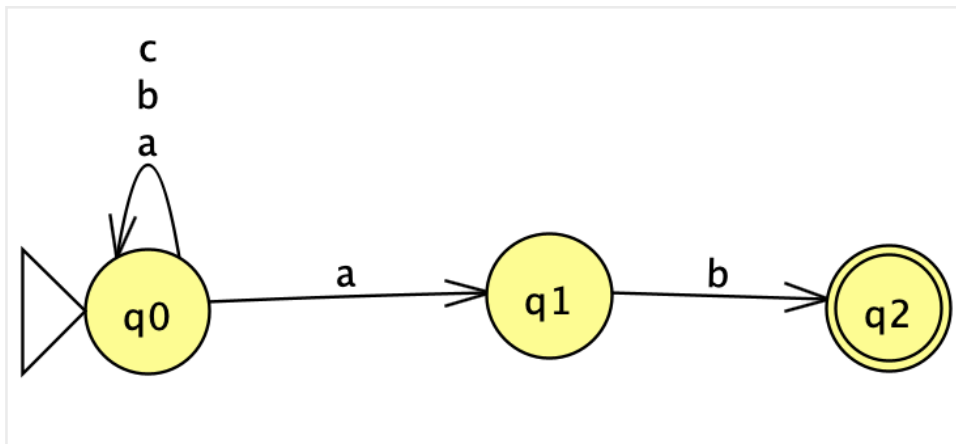


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3. Cadeias terminando em ab

ER: $(a|b|c)^*ab$

AF:

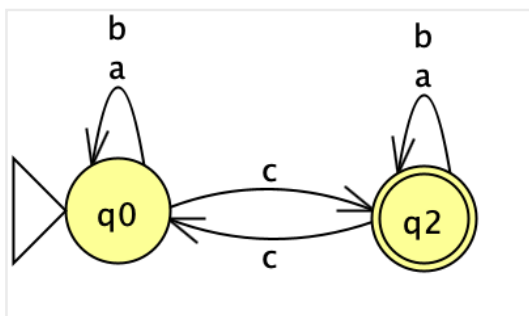


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4. Cadeias com quantidade impar de símbolos c

ER: $((a|b)^*c(a|b)^*c)^*(a|b)^*c(a|b)^*$

AF:

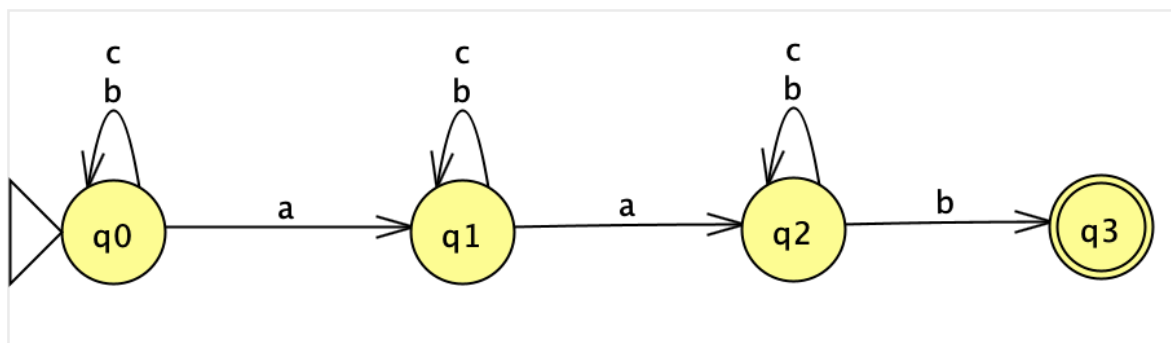


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5. Cadeias tendo 2 a e terminando com b

ER: $(b|c)^*a(b|c)^*a(b|c)^*b$

AF:



Gabarito Geral Questões Livro +

1) Cadeias de comprimento qualquer, incluindo zero;

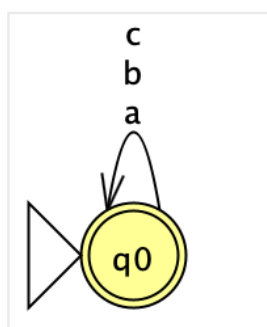
Gramática:

LHS		RHS
S	→	aS
S	→	bS
S	→	cS
S	→	λ

ER: $(a|b|c)^*$

Teste: $\{\epsilon, a, b, c, aa, ab, ac, ba, bb, bc, ca, cb, cc, aaa, aaab\}$

AF:



2) Cadeias de comprimento qualquer, maior que zero

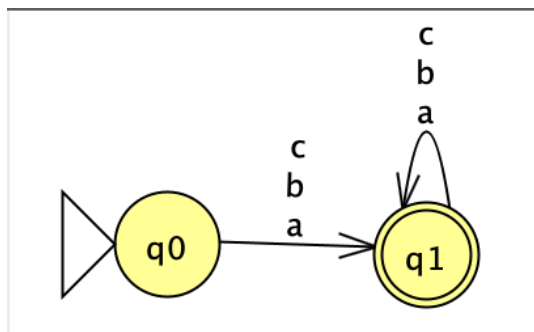
Gramática:

LHS		RHS
S	$\rightarrow$	aS
S	$\rightarrow$	bS
S	$\rightarrow$	a
S	$\rightarrow$	b
S	$\rightarrow$	c
S	$\rightarrow$	cS
S	$\rightarrow$	λ

ER: $(a|b|c)(a|b|c)^*$

Teste: {a, b, c, aa, ab, ac, ba, bb, bc, ca, cb, cc, aaa, aaab}

AF:



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3) Cadeias de comprimento 3

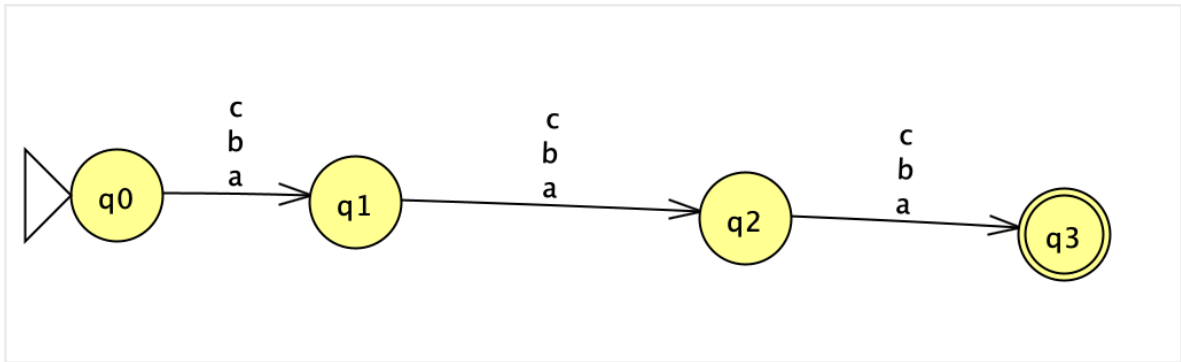
Gramática:

LHS		RHS
S	$\rightarrow$	AAA
A	$\rightarrow$	a
A	$\rightarrow$	b
A	$\rightarrow$	c

ER: $(a|b|c)(a|b|c)(a|b|c)$

Teste: {bca, aab, aca, bab, cab, acc, abb, abc, acb, aaa, cbb, baa}

AF:



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4) Cadeias de comprimento diferente de 3

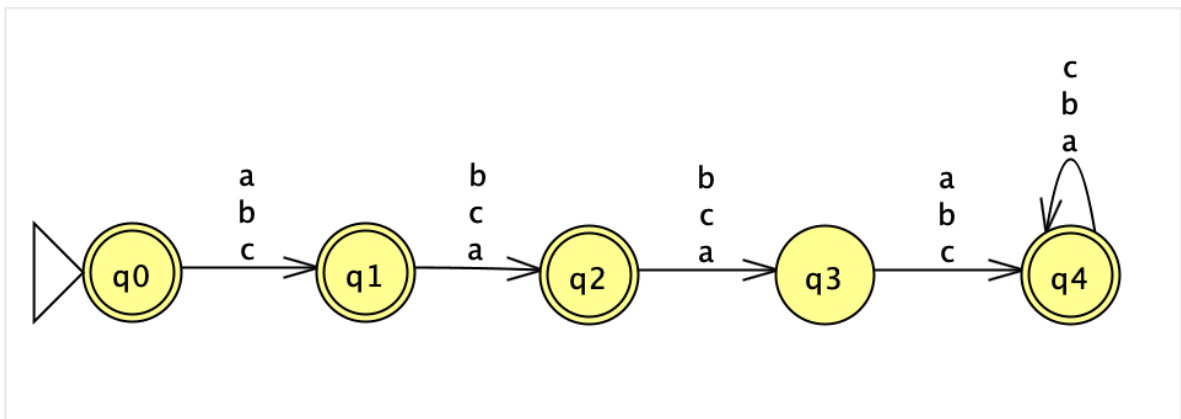
Gramática:

LHS		RHS
S	→	A
S	→	AA
S	→	AAAAB
A	→	a
A	→	b
A	→	c
B	→	AB
B	→	λ

ER: $\epsilon|a|b|c|(a|b|c)(a|b|c)|(a|b|c)(a|b|c)(a|b|c)(a|b|c)(a|b|c)^*$

Teste: {a, bc, bbcc, bcabaab, bcaa, c, ϵ , acababab, acaacabbab, cabacacb, aabc, babac}

AF:



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5) Cadeias de comprimento maior que 3

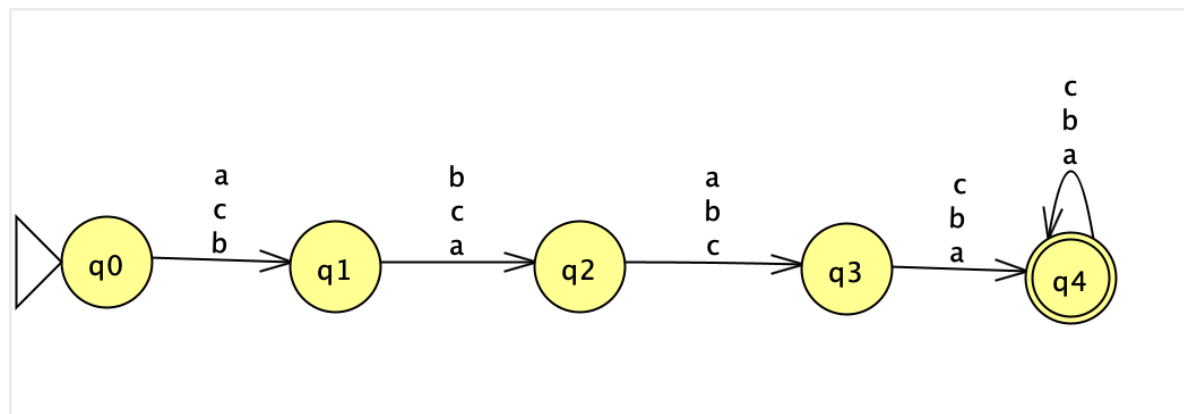
Gramática:

LHS		RHS
S	→	AAAAB
A	→	a
A	→	b
A	→	c
B	→	AB
B	→	λ

ER: $(a|b|c)(a|b|c)(a|b|c)(a|b|c)(a|b|c)^*$

Teste: {bbcc, bcabaab, bcaa, cababab, acaacabbab, cabacacb, aabc, babac, abaaa, bbcb}

AF:



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6) Cadeias de comprimento maior ou igual a 3

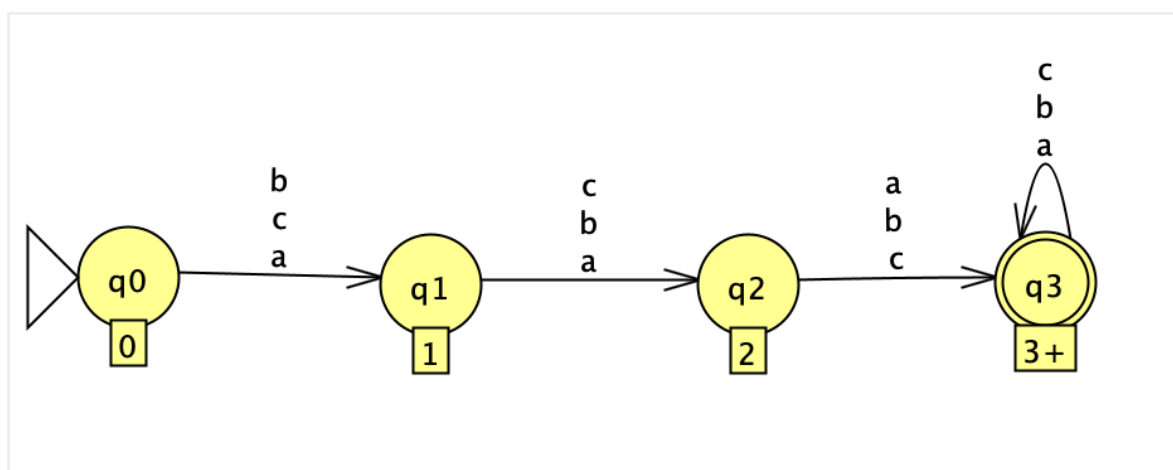
Gramática:

LHS		RHS
S	→	AAAB
A	→	a
A	→	b
A	→	c
B	→	AB
B	→	λ

ER: $(a|b|c)(a|b|c)(a|b|c)(a|b|c)^*$

Teste: {bbc, bcabaab, bcaa, cababab, acaacabbab, cabacacb, aabc, babac, abaaa, bcb}

AF:



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7) Cadeias de comprimento menor que 3

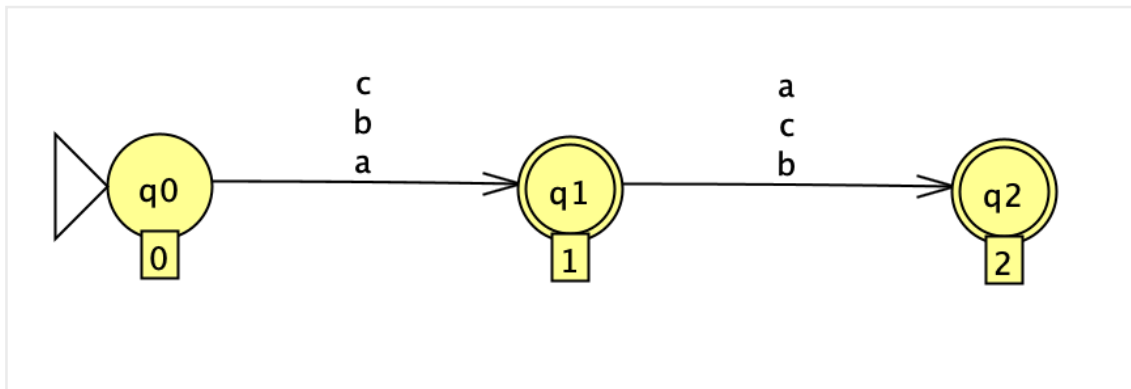
Gramática:

LHS		RHS
S	→	AA
A	→	a
A	→	b
A	→	c
A	→	λ

ER: $\epsilon|a|b|c|(a|b|c)(a|b|c)$

Teste: { ϵ , a, b, c, aa, ab, ac, ba, bb, bc, ca, cb, cc}

AF:



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8) Cadeias de comprimento múltiplo de 3

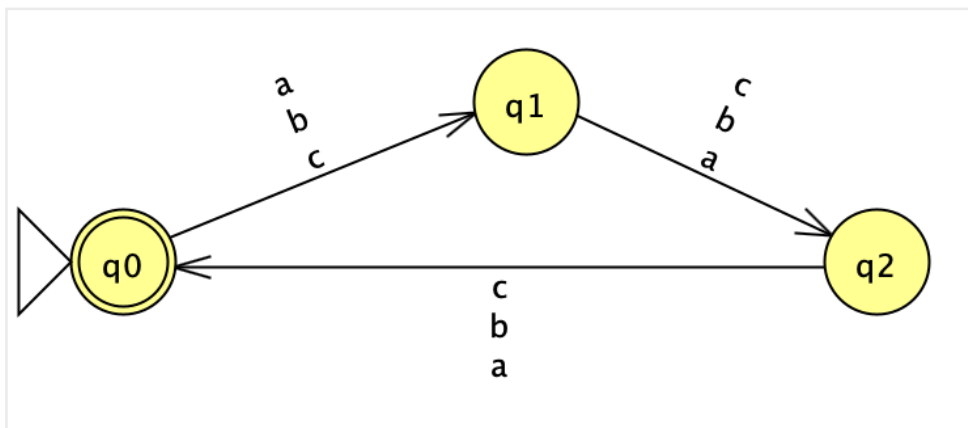
Gramática:

LHS		RHS
S	→	AAAS
A	→	a
A	→	b
A	→	c
S	→	λ

ER: $((a|b|c)(a|b|c)(a|b|c))^*$

Teste: {bca, acabababb, ϵ , acabab, cabacacbb, aabcbabaccba, baaaba, aaa, bbbbbb}

AF:



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9) Cadeias de comprimento múltiplo de 4

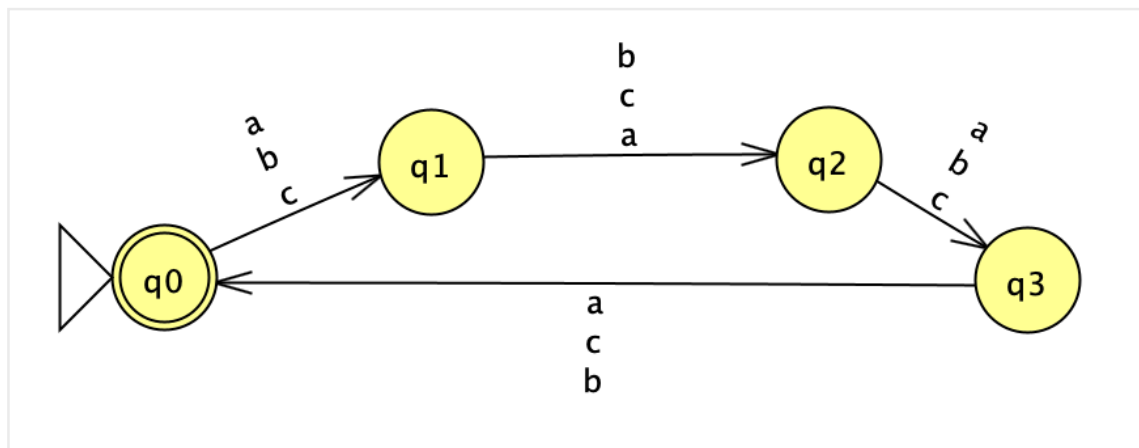
Gramática:

LHS		RHS
S	$\rightarrow$	AAAAS
A	$\rightarrow$	a
A	$\rightarrow$	b
A	$\rightarrow$	c
S	$\rightarrow$	λ

ER: $((a|b|c)(a|b|c)(a|b|c)(a|b|c))^*$

Teste: {tbcaa, acababab, ϵ , aabcbabaccba, aaac}

AF:



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10) Cadeias de comprimento com uma quantidade par de símbolos

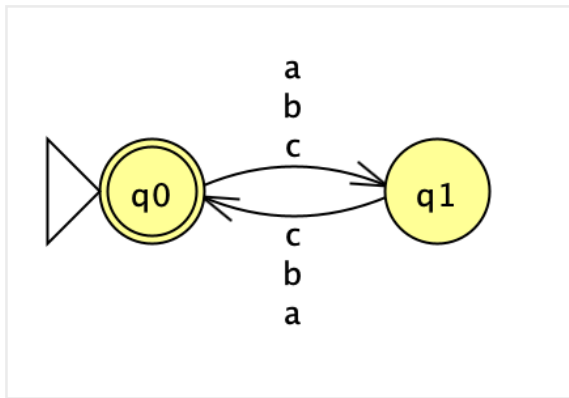
Gramática:

LHS		RHS
S	$\rightarrow$	AAS
A	$\rightarrow$	a
A	$\rightarrow$	b
A	$\rightarrow$	c
S	$\rightarrow$	λ

ER: $((a|b|c)(a|b|c))^*$

Teste: { ϵ , bb, ac, aabc, abac, abbc, abcc, acac, acbc, aaaacb, bababc}

AF:



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11) Cadeias de comprimento com uma quantidade ímpar de símbolos

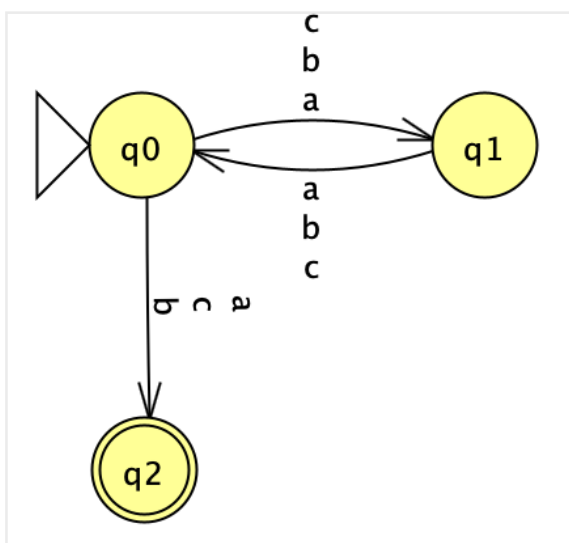
Gramática:

LHS		RHS
S	→	AAS
A	→	a
A	→	b
A	→	c
S	→	A

ER: $(a|b|c)((a|b|c)(a|b|c))^*$

Teste: {bcb, acbbb, a, c, aabcbbbb, bbbacbbba, abc, cbabc, aaa}

AF:



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12) Cadeias se iniciando com a

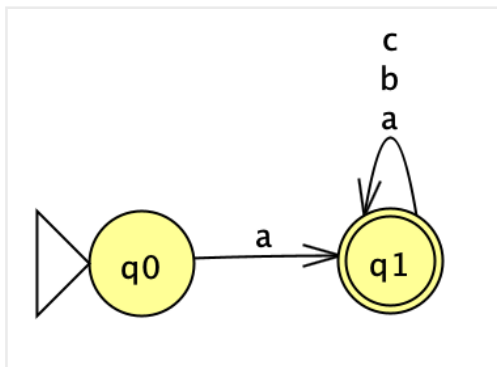
Gramática:

LHS		RHS
S	→	aA
A	→	aA
A	→	bA
A	→	cA
A	→	λ

ER: $a(a|b|c)^*$

Teste: {a, ab, aa, aaaa, acc, acbba}

AF:



13) Cadeias de iniciando com abb

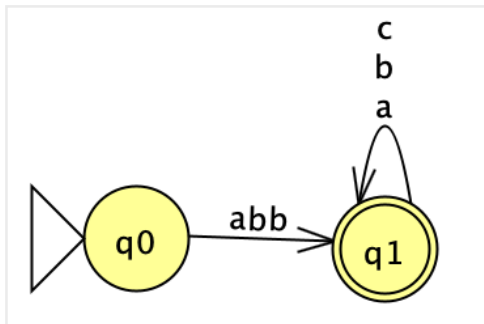
Gramática:

LHS		RHS
S	→	aabA
A	→	aA
A	→	bA
A	→	cA
A	→	λ

ER: $abb(a|b|c)^*$

Teste: {abb, abba, abbabb, abbcabbc, abbccbbb}

AF:



14) Cadeias que não iniciam com a

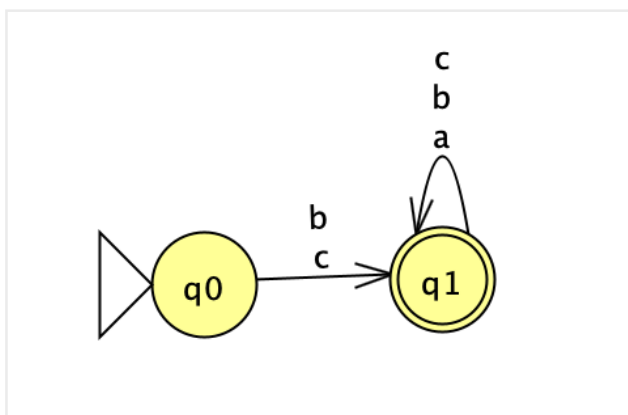
Gramática:

LHS		RHS
S	→	bA
A	→	aA
A	→	bA
A	→	cA
A	→	λ
S	→	cA
S	→	λ

ER: $\epsilon | b(a|b|c)^* | c(a|b|c)^*$

Teste: {bb, cba, bcbca, bacc, cbba}

AF:



15) Cadeias que não iniciam com aa

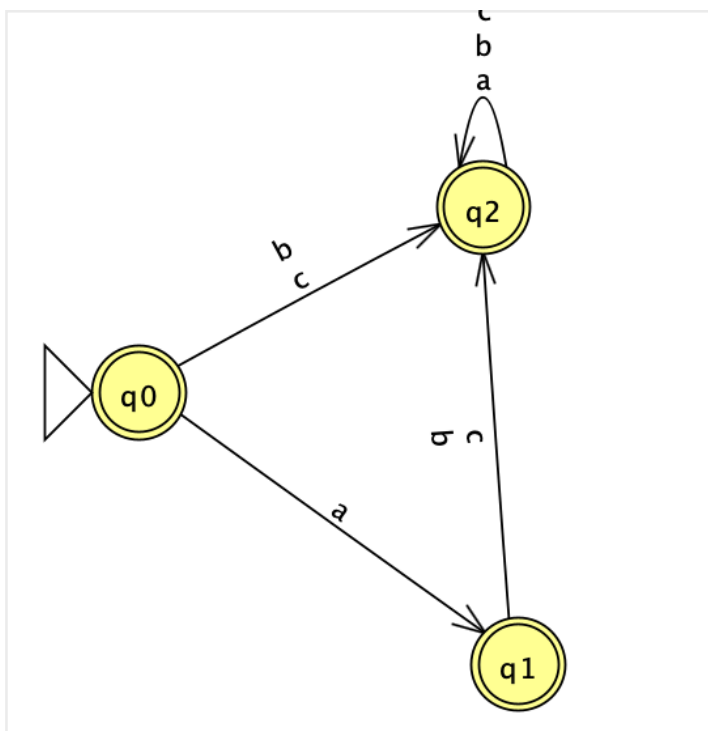
Gramática:

LHS		RHS
S	→	abA
A	→	aA
A	→	bA
A	→	cA
A	→	λ
S	→	acA
S	→	λ
S	→	a
S	→	bA
S	→	cA

ER: $\epsilon|a|b(a|b|c)^*|c(a|b|c)^*|ab(a|b|c)^*|ac(a|b|c)^*$

Teste: {abb, aba, abbb, bbabb, bcabbc, babbccc}

AF:



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16) Cadeias terminando com pelo menos três símbolos b consecutivos

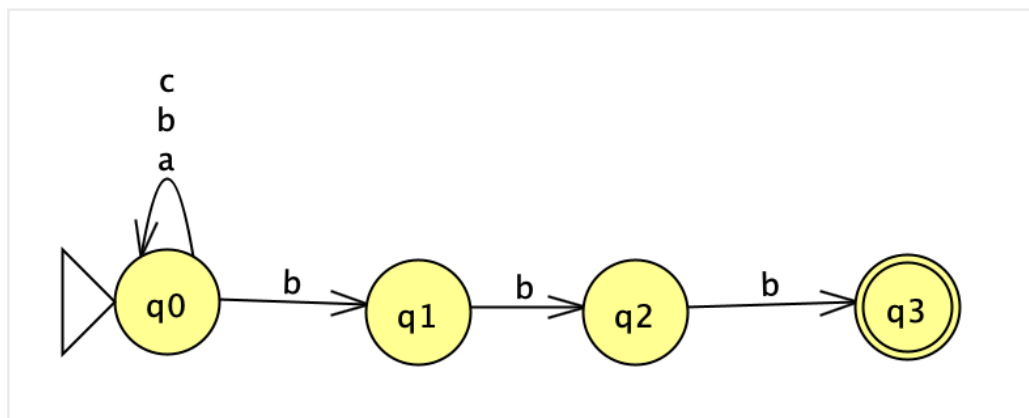
Gramática:

LHS		RHS
S	$\rightarrow$	Abbb
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ

ER: $(a|b|c)^*bbb$

Teste: {bbb, acbbb, aabcbbb, abacbbb, abbb, bbbb, acacbbb, bbbacbbb, abababbbb}

AF:



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17) Cadeias terminando com pelo três símbolos b consecutivos, e não mais que isso

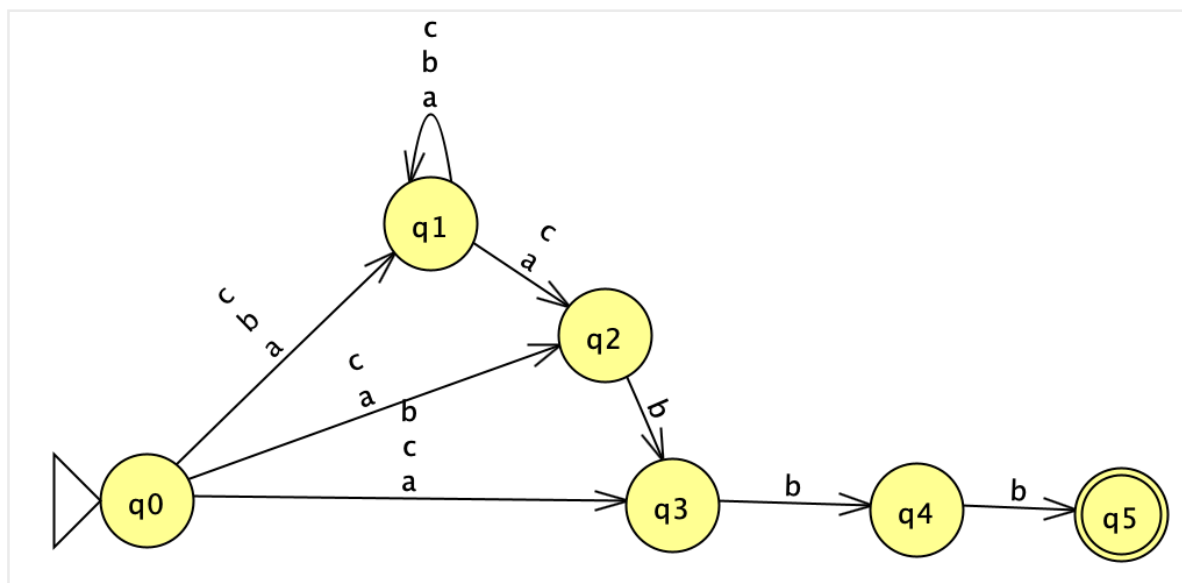
Gramática:

LHS		RHS
S	→	Abbb
A	→	aA
A	→	bA
A	→	cA
A	→	a
A	→	c
S	→	bbb

ER: $bbb|(a|b|c)^*(a|c)bbb$

Teste: {bbb, acbbb, aabcbbb, abacbbb, abbb, acacbbb, bbbacbbb, abababbb}

AF:



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18) Cadeias que não terminam com dois símbolos b consecutivos

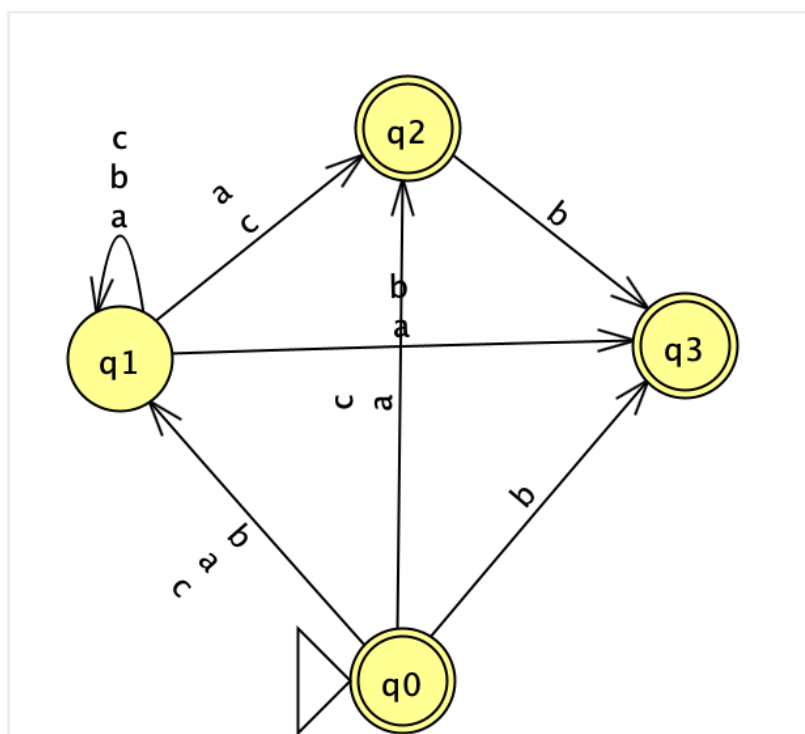
Gramática:

LHS		RHS
S	→	Aa
A	→	aA
A	→	bA
A	→	cA
S	→	b
S	→	λ
S	→	Ac
S	→	Aab
S	→	Acb
A	→	λ

ER: $\epsilon|a|b|c|(a|b|c)^*(a|c)|(a|b|c)^*(a|c)b$

Teste: {a, b, c, acbba, aab, abacbbc, abcc, acacbc, bbbacaa, ababa, ccc}

AF:



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19) Cadeias iniciando com a e terminando com c

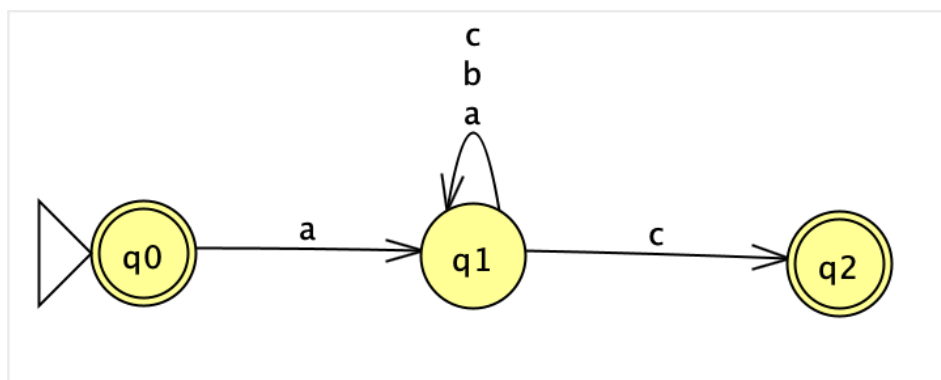
Gramática:

LHS		RHS
S	→	aAc
A	→	aA
A	→	bA
A	→	cA
A	→	λ

ER: $ac|a(a|b|c)^*c$

Teste: {tac, abc, acc, aac, aabc, abac, abbc, abcc, acac, acbc, aaaac, aaabc}

AF:



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20) Cadeias que iniciam com a e não terminam com c

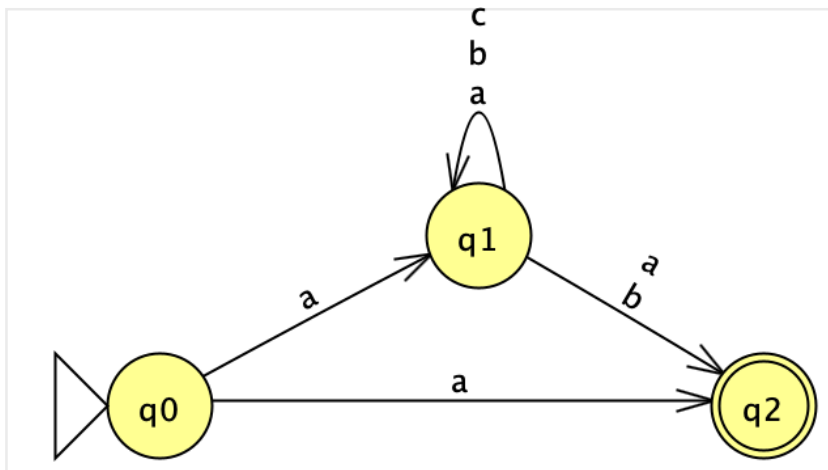
Gramática:

LHS		RHS
S	→	aAa
A	→	aA
A	→	bA
A	→	cA
A	→	λ
S	→	aAb
S	→	a

ER: $a|a(a|b|c)^*(a|b)$

Teste: {a, ab, acb, aaca, aabcb, aba, abb, abcca, acaca, acb, aaaa, aaaab}

AF:



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21) Cadeias que não iniciam com a e terminam com c

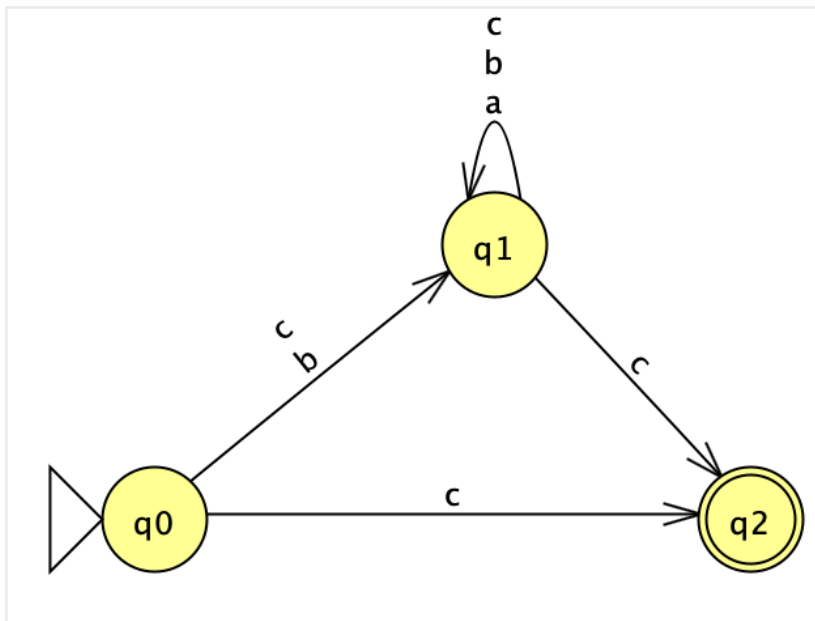
Gramática:

LHS		RHS
S	$\rightarrow$	bAc
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ
S	$\rightarrow$	cAc
S	$\rightarrow$	c

ER: $c|(b|c)(a|b|c)^*c$

Teste: {c, bc, bac, bbc, ccc, babcb, babac, babbc, bbcc, cacacac, cbc, ccccc, bbbbc}

AF:



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22) Cadeias que não iniciam com a e não terminam com c

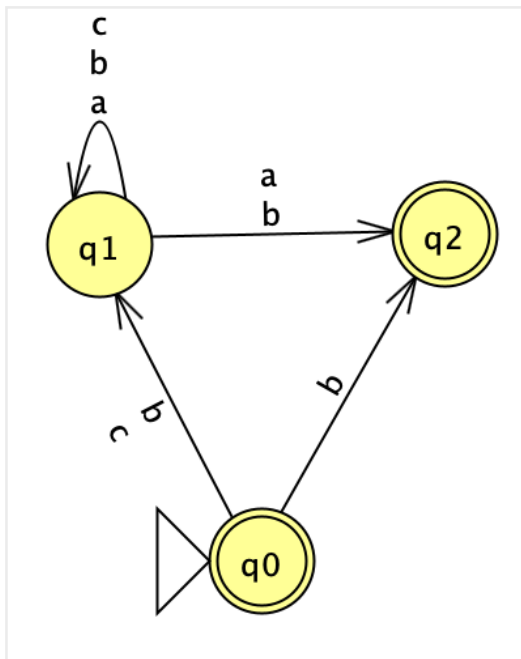
Gramática:

LHS		RHS
S	$\rightarrow$	BAC
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ
S	$\rightarrow$	b
S	$\rightarrow$	λ
B	$\rightarrow$	b
B	$\rightarrow$	c
C	$\rightarrow$	a
C	$\rightarrow$	b

ER: $\epsilon|b|c|(b|c)(a|b|c)^*(a|b)$

Teste: {baca, cb, cacb, caaa, caabcb, babacb, babbcbb, cabccb, ca, bb, bcab, bbbcb}

AF:



23) Cadeias com exatamente três símbolos b

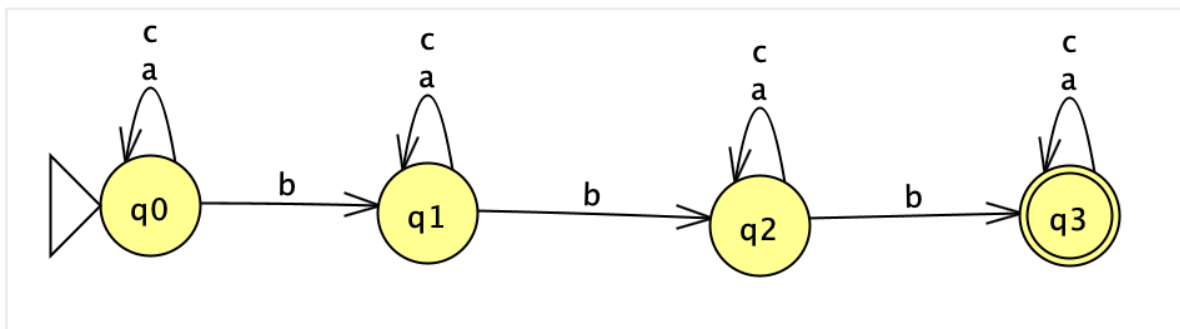
Gramática:

LHS		RHS
S	$\rightarrow$	AbAbAbA
A	$\rightarrow$	aA
A	$\rightarrow$	cA
A	$\rightarrow$	λ

ER: $(a|c)^*b(a|c)^*b(a|c)^*b(a|c)^*$

Teste: {bcbbbb, acbbb, bbab, cbbb, aabcbb, bacabccba, bbbc, cbabcba, ababab}

AF:



24) Cadeias com pelo menos dois símbolos a

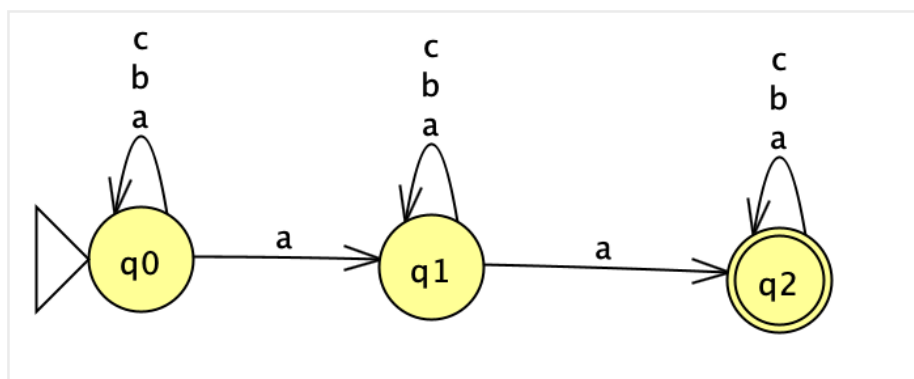
Gramática:

LHS		RHS
S	→	AaAaA
A	→	aA
A	→	cA
A	→	λ
A	→	bA

ER: $(b|c)^*a(b|c)^*a(a|b|c)^*$

Teste: {bcbaab, acbabab, aaabbab, cababab, aabcbabaa, bacabcaacba, aaabaaabbc, aa}

AF:



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25) Cadeias com no máximo quatro símbolos c

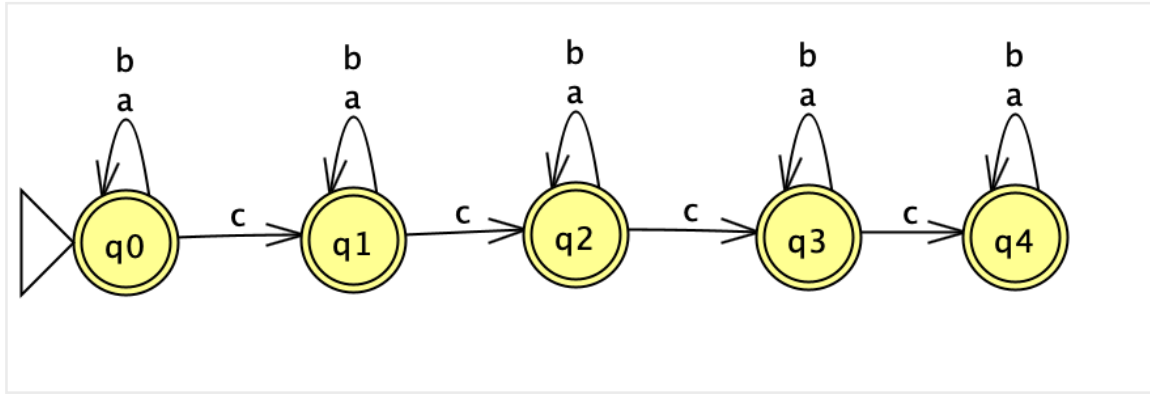
Gramática:

LHS		RHS
S	→	ABABABABA
A	→	aA
A	→	bA
A	→	λ
B	→	λ
B	→	c

ER: $(a|b)^*(c|ε)(a|b)^*(c|ε)(a|b)^*(c|ε)(a|b)^*(c|ε)(a|b)^*$

Teste: {bcbaab, acbabab, ccabbab, cabacacb, aabcbabac, baabaaba, aaabbc, ccc}

AF:



26) Cadeias que contenham no mínimo dois símbolos a ou no máximo três símbolos c

Gramática:

LHS		RHS
S	→	AaAaA
A	→	aA
A	→	bA
A	→	λ
A	→	cA
S	→	BcBcBcB
B	→	aB
B	→	bB
B	→	λ

ER: $((b|c)^*a(b|c)^*a(a|b|c)^*)((a|b)^*(c|e)(a|b)^*(c|e)(a|b)^*(c|e)(a|b)^*)$

Teste: {abccabc, abaccbcb, aaabcc, acccbc, abcabcabc, cababc, aa, ababbabca, ccc}

AF: Não Consegui (Gabarito do Livro!)

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27) Cadeias com no mínimo três e no máximo cinco símbolos a

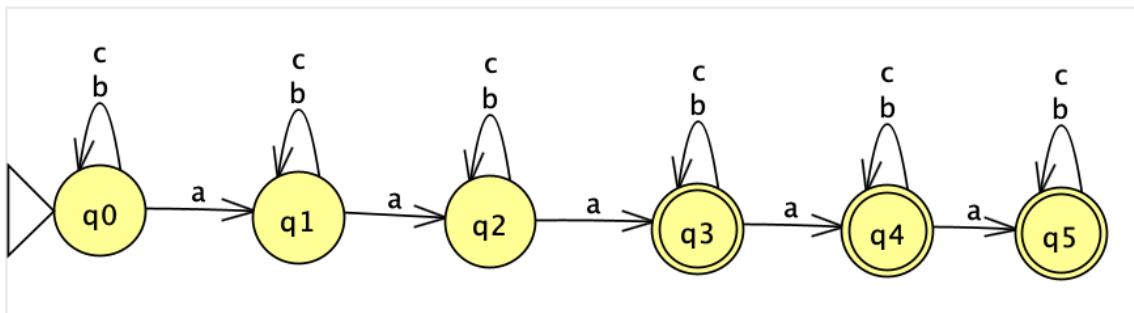
Gramática:

LHS		RHS
S	$\rightarrow$	AaAaAaABABA
A	$\rightarrow$	cA
A	$\rightarrow$	bA
A	$\rightarrow$	λ
B	$\rightarrow$	a
B	$\rightarrow$	λ

ER: $((b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*)|((b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*)|((b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*a(b|c)^*)$

Teste: {bcabaab, acababab, acaacabbab, cabacacb, aabcbabac, baaba, aaabbc}

AF:



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28) Cadeias que iniciam e terminam com símbolos diferentes

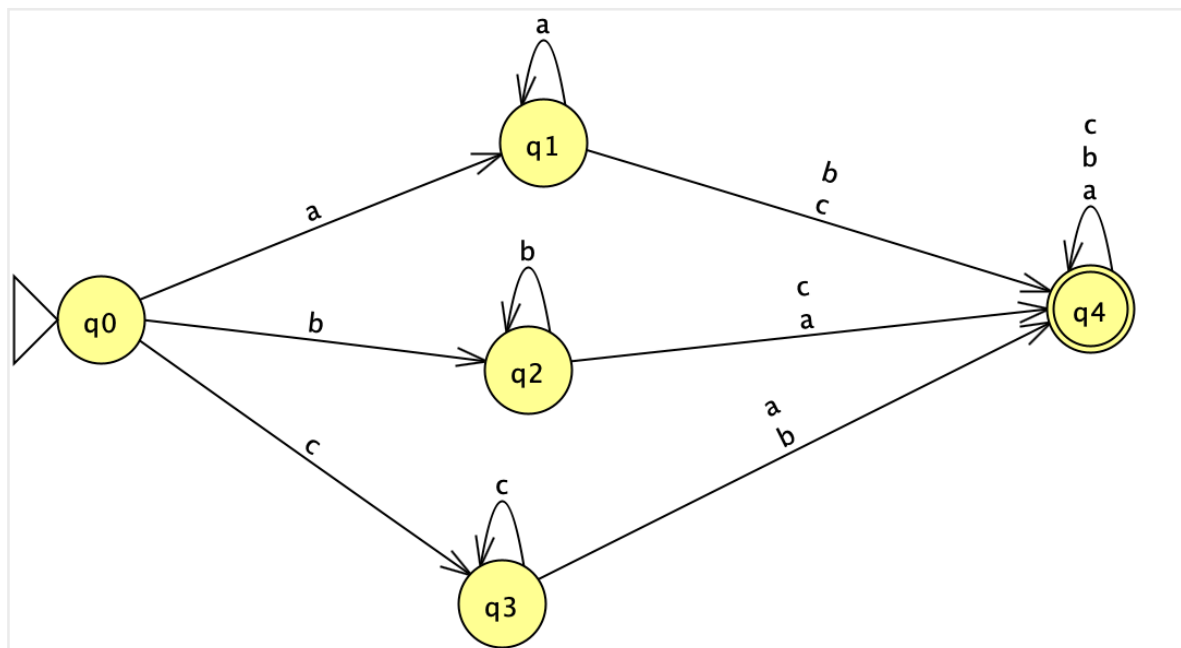
Gramática:

LHS		RHS
S	→	aAb
A	→	aA
A	→	bA
A	→	cA
A	→	λ
S	→	aAc
S	→	bAa
S	→	bAc
S	→	cAa
S	→	cAb

ER: $a(a|b|c)^*(b|c)|b(a|b|c)^*(a|c)|c(a|b|c)^*(a|b)$

Teste: {abccabc, abaccbcb, caabca, acccb, abc, bababc, ba, bacacc, bcbabca, cca}

AF:



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29) Cadeias em que os símbolos aparecem estritamente na ordem: primeiro os símbolos a (zero ou mais), depois os símbolos b (idem) e por último os símbolos c (idem)

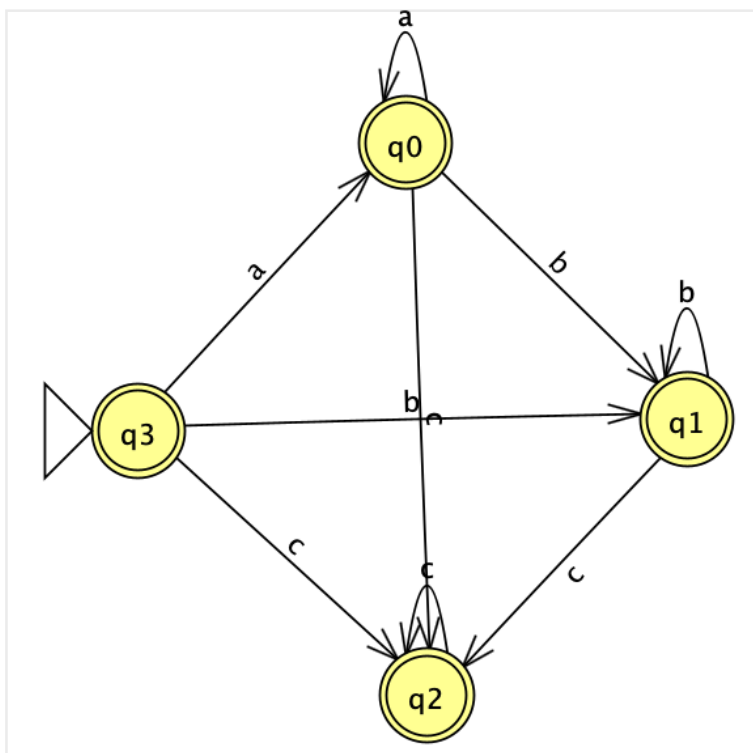
Gramática:

LHS		RHS
S	$\rightarrow$	aS
A	$\rightarrow$	bA
A	$\rightarrow$	B
A	$\rightarrow$	λ
S	$\rightarrow$	A
B	$\rightarrow$	cB
B	$\rightarrow$	λ

ER: $a^*b^*c^*$

Teste: {abcc, abbbbbb, cccc, aabbcc, abc, bbbc, b, aaa, aaccc, bc, ϵ , abc}

AF:



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30) Cadeias que possuem uma sequência de um ou mais símbolos b imediatamente à direita de cada símbolo a

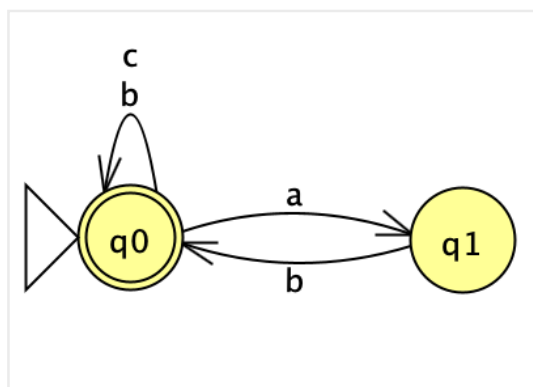
Gramática:

LHS		RHS
S	$\rightarrow$	AS
S	$\rightarrow$	λ
A	$\rightarrow$	b
A	$\rightarrow$	c
A	$\rightarrow$	abB
B	$\rightarrow$	λ
B	$\rightarrow$	bB

ER: $(b|c|ab(b)^*)^*$

Teste: {abccabc, abbabbbccbc, cabcabb, abcccb, abc, bababc, b, babcabbcc, bbcab}

AF:



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31) Cadeias que não contenham símbolos b justapostos

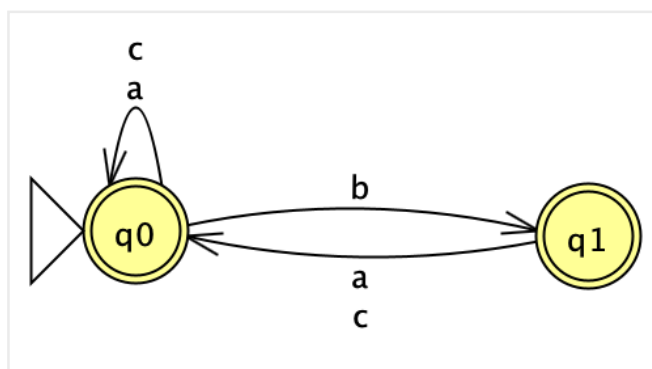
Gramática:

LHS		RHS
S	$\rightarrow$	aS
S	$\rightarrow$	λ
A	$\rightarrow$	aS
A	$\rightarrow$	cS
A	$\rightarrow$	λ
S	$\rightarrow$	cS
S	$\rightarrow$	bA

ER: $(a|c|b(a|c))^*(b|\epsilon)$

Teste: $\{\epsilon, a, c, ab, cb, abc, acb, bab, bcab, cac\}$

AF:



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32) Cadeias que não contenham símbolos a a direita em qualquer posição de símbolos c

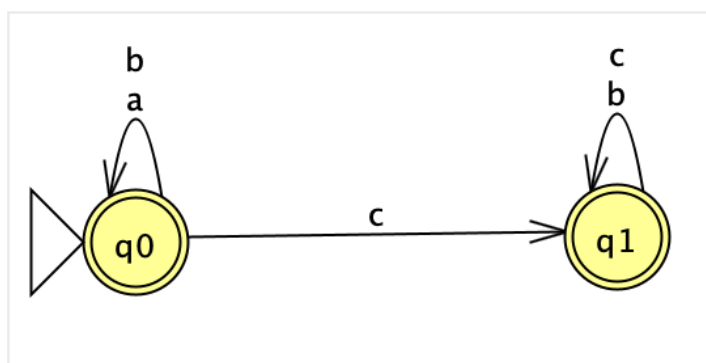
Gramática:

LHS		RHS
S	$\rightarrow$	aS
S	$\rightarrow$	λ
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ
S	$\rightarrow$	bS
S	$\rightarrow$	cA

ER: $(a|b)^*(c|b)^*$

Teste: $\{\epsilon, a, b, c, aa, bb, cc, ab, bc, aabcc, abbc\}$

AF:



===== 33) Cadeias com quantidade par de símbolos b

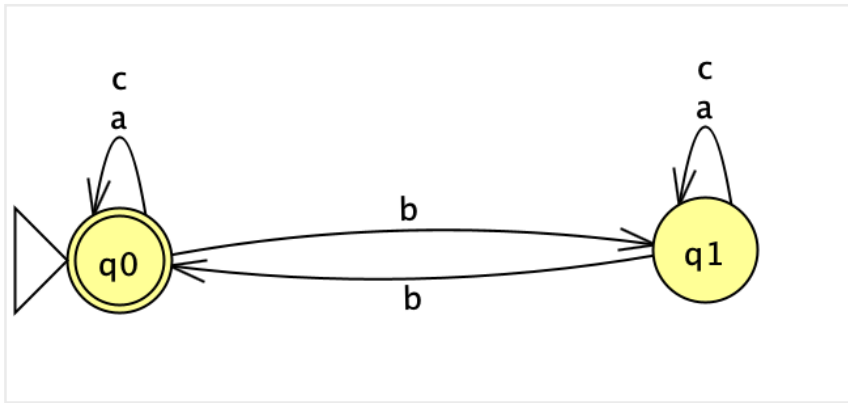
Gramática:

**

ER: $(a|c)^*(b(a|c)^*b(a|c)^*)^*$

Teste: $\{\epsilon, a, c, bb, abb, bba, bcb, abcba, bbbb\}$

AF:



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34) Cadeias com quantidade impar de símbolos c

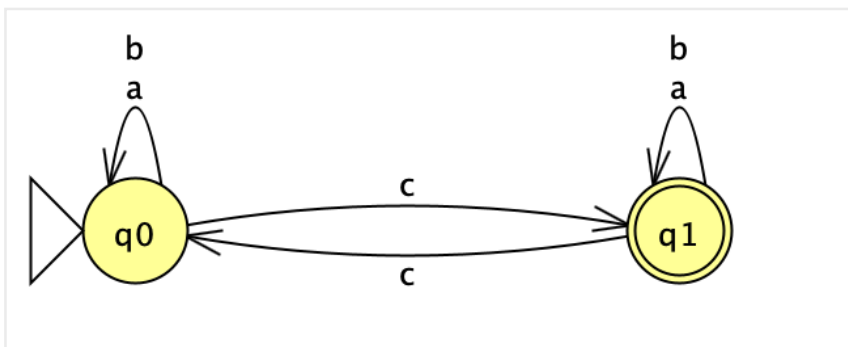
Gramática:

LHS		RHS
S	→	AB
A	→	CcCcA
A	→	λ
B	→	CcC
C	→	aC
C	→	bC
C	→	λ

ER: $(a|b)^*c(a|b)^*(c(a|b)^*c(a|b)^*)^*$

Teste: {c, ac, bc, acca, ccc, abca, bcbcc, ccccc}

AF:



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35) Cadeias com quantidade par de símbolos a e impar de símbolos c

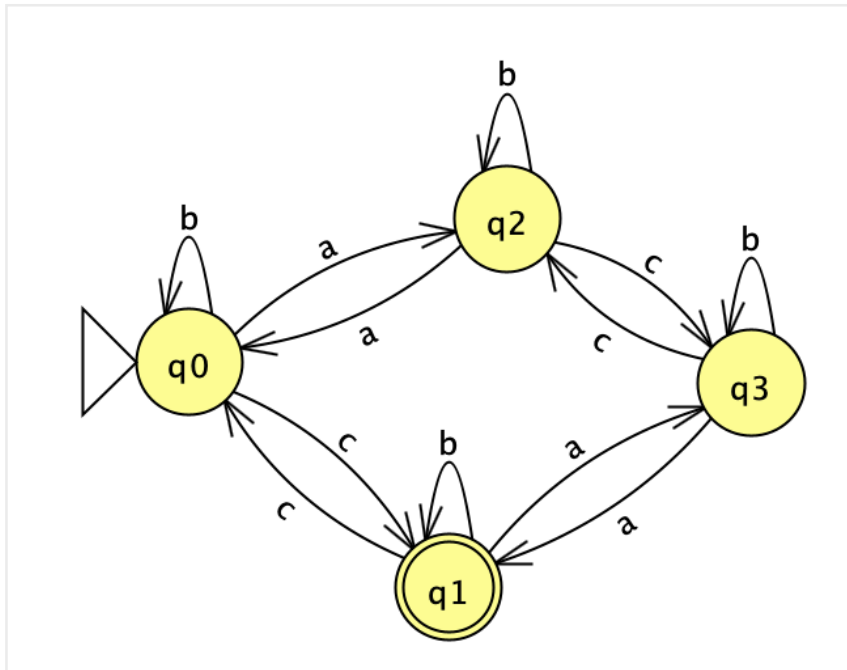
Gramática:

LHS		RHS
S	→	aB
S	→	bS
S	→	cA
A	→	λ
A	→	aC
A	→	bA
A	→	cS
B	→	aS
B	→	bB
B	→	cC
C	→	aA
C	→	bC
C	→	cB

ER: $(b|aa|ac(cc)^*ca|ac(cc)^*b(bb)^*b(cc)^*ca)^*c(cc)^*(b|aa)^*$

Teste: {cabccabcc, aaacacbcb, bccc, cb, aabcabacaabc, cabccabcc, accca, bacacc, aca,}

AF:



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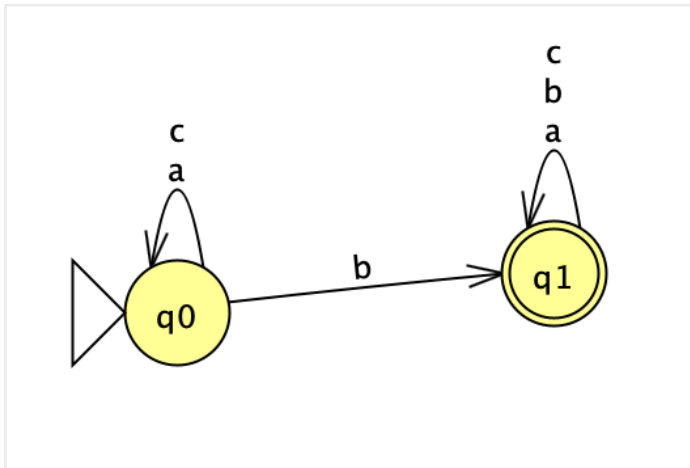
36) Cadeias que contenham o símbolo b
 Gramática:

LHS		RHS
S	$\rightarrow$	AbA
A	$\rightarrow$	aA
A	$\rightarrow$	bA
A	$\rightarrow$	cA
A	$\rightarrow$	λ

ER: $(a|b|c)^*b(a|b|c)^*$

Teste: {b, ab, ba, cbc, aaab, cccbccc, abc, bbb}

AF:



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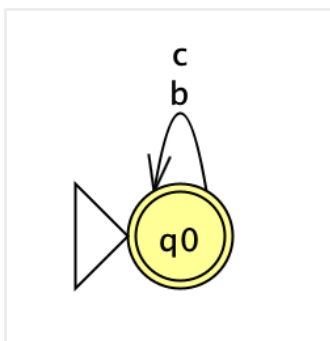
37) Cadeias que não contenham símbolo a
 Gramática:

LHS		RHS
S	$\rightarrow$	bS
S	$\rightarrow$	cS
S	$\rightarrow$	λ

ER: $(b|c)^*$

Teste: $\{\epsilon, b, c, bb, cc, bc, cb, bbb, ccc, bbcbc\}$

AF:



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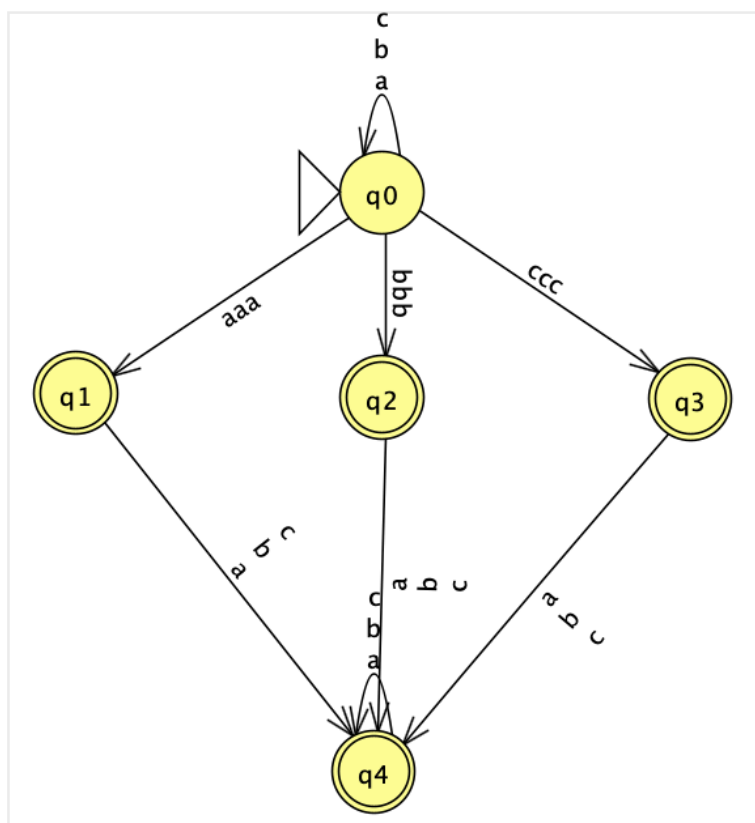
38) cadeias que contenham pelo menos 3 símbolos iguais consecutivos
 Gramática:

LHS		RHS
S	→	ABA
A	→	aA
A	→	bA
A	→	cA
A	→	λ
B	→	aaa
B	→	bbb
B	→	ccc

ER: $(a|b|c)^*(aaa|bbb|ccc)(a|b|c)^*$

Teste: {aaa, bbb, ccc, aaaa, bbbba, accccb, abaaa, cccb}

AF: Não consegui



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39) Cadeias que contenham 2 símbolos iguais consecutivos

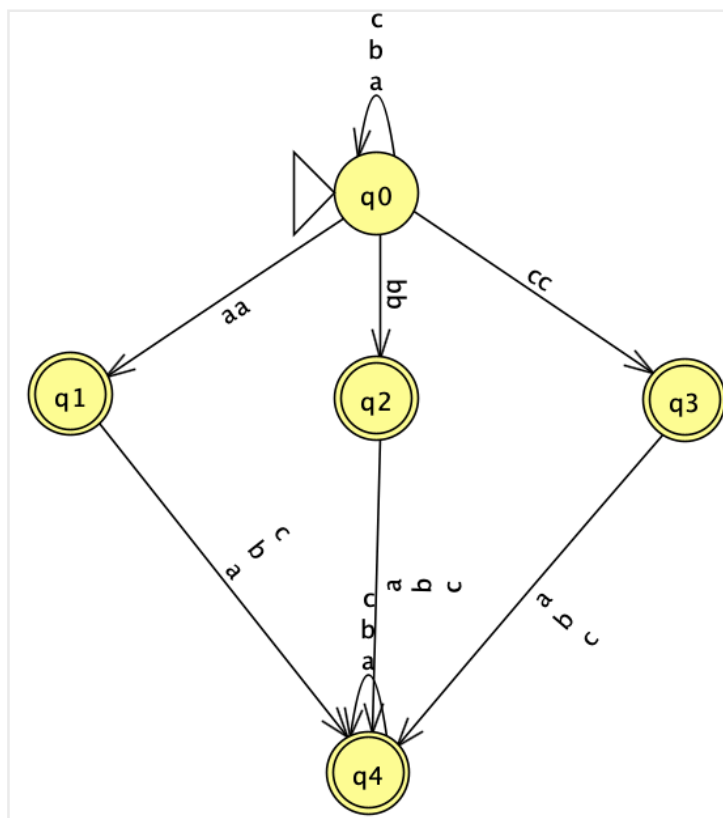
Gramática:

LHS		RHS
S	→	ABA
A	→	aA
A	→	bA
A	→	cA
A	→	λ
B	→	aa
B	→	bb
B	→	cc

ER: $(a|b|c)^*(aa|bb|cc)(a|b|c)^*$

Teste: {aa, bb, cc, aab, bcc, abba, accb, baab, ccc}

AF:



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40) Cadeias que contenham a subcadeia abc

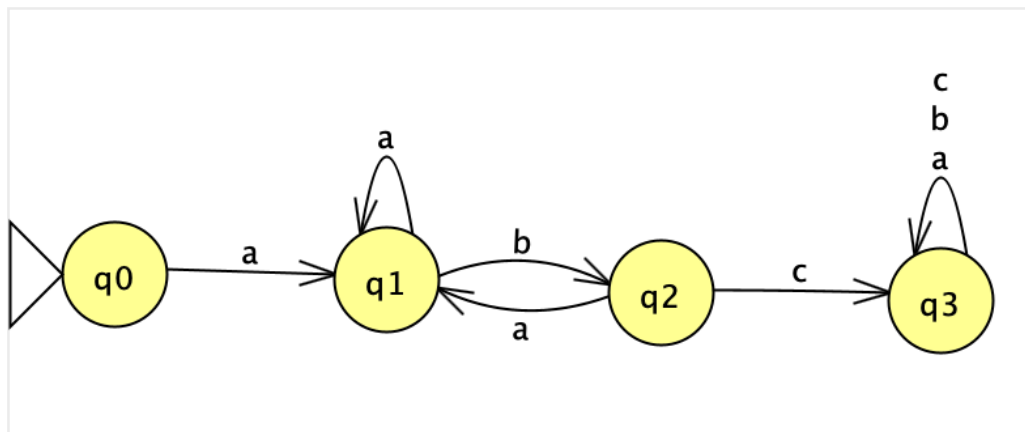
Gramática:

LHS		RHS
S	→	AabcA
A	→	aA
A	→	bA
A	→	cA
A	→	λ

ER: $(a|b|c)^*abc(a|b|c)^*$

Teste: {abc, aabc, abcc, babcb, cabca, aaabcc, bcabc}

AF:



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41) Cadeias que não contenham a subcadeia ab

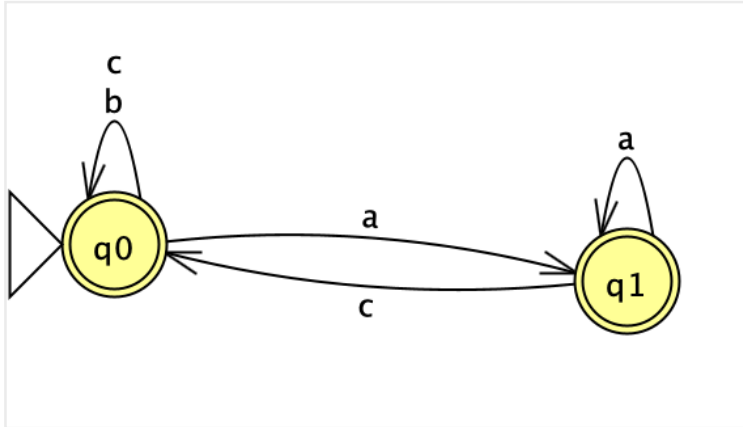
Gramática:

LHS		RHS
S	→	bS
A	→	aA
A	→	cS
A	→	λ
S	→	cS
S	→	λ
S	→	aA

ER: $(b|c)^*(a(a)^*(c))^*(a)^*$

Teste: {ε, a, b, c, aa, ac, ba, ca, bb, cc, cba, aca}

AF:



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42) Cadeias que não contenham subcadeia abc

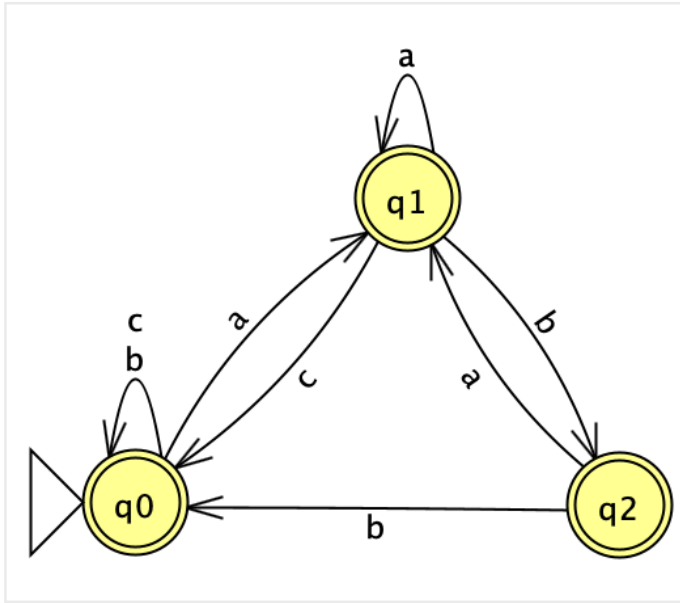
Gramática:

LHS		RHS
S	→	bS
A	→	aA
A	→	cS
A	→	bB
A	→	λ
S	→	cS
S	→	λ
S	→	aA
B	→	aA
B	→	bB
B	→	λ

ER: $(b|c|a|ab|ab(a|b|c)^*c)^*$

Teste: {ε, a, b, c, ab, ba, ac, bc, cb, aab, abba, acc}

AF:



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